MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746273
COMPUTER-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Computer-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746273	Circuit Card Assembly, Computer-3.

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Definition.- This assembly is the third of six single ended printed circuit boards consisting of operational amplifiers and multiplex switches necessary to provide various mathematical parameters used in the analog computer section of the STG.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746273.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +5 VDC
b. +5 VDC REF
c. -5 VDC REF
d. +10 VDC
e. -10 VDC
f. +15 VDC
g. -15 VDC

3.2.1.2 Multplex Switch Outputs.- When BCD inputs are applied to the multiplex switches, the outputs shall be incremental voltages.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746273 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746273 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

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4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (See Figure 1)

4.2.1.1 Power Supply (PS 2 & PS 3).- The power supply shall be 10 VDC ± 1 mV, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS 4 & PS5).- The power supply shall be 15 VDC ± 1 mV, 100 mA or greater and a ripple of less than 10 mV RMS,

4.2.1.3 Power Supply (PS 6 & PS7).- The power supply shall be 20 VDC ± 1 mV, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.4 Dummy Load.- The dummy loads shall be 10K ohm, ¼ watt, 0.1% resistors.

4.2.1.5 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltages scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standrad ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +5.000
b. +5.000
c. -5.000
d. +10.000 ± 0.001 VDC
e. -10.000 ± 0.001 VDC
f. +15.000 ± 0.001 VDC
g. -15.000 ± 0.001 VDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-Up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Reference Voltage Test.- When power is connected to the circuit board as shown in Figure 1, the voltages on pins 1 and 2 shall be plus (+) and minus (-) five volts respectively.

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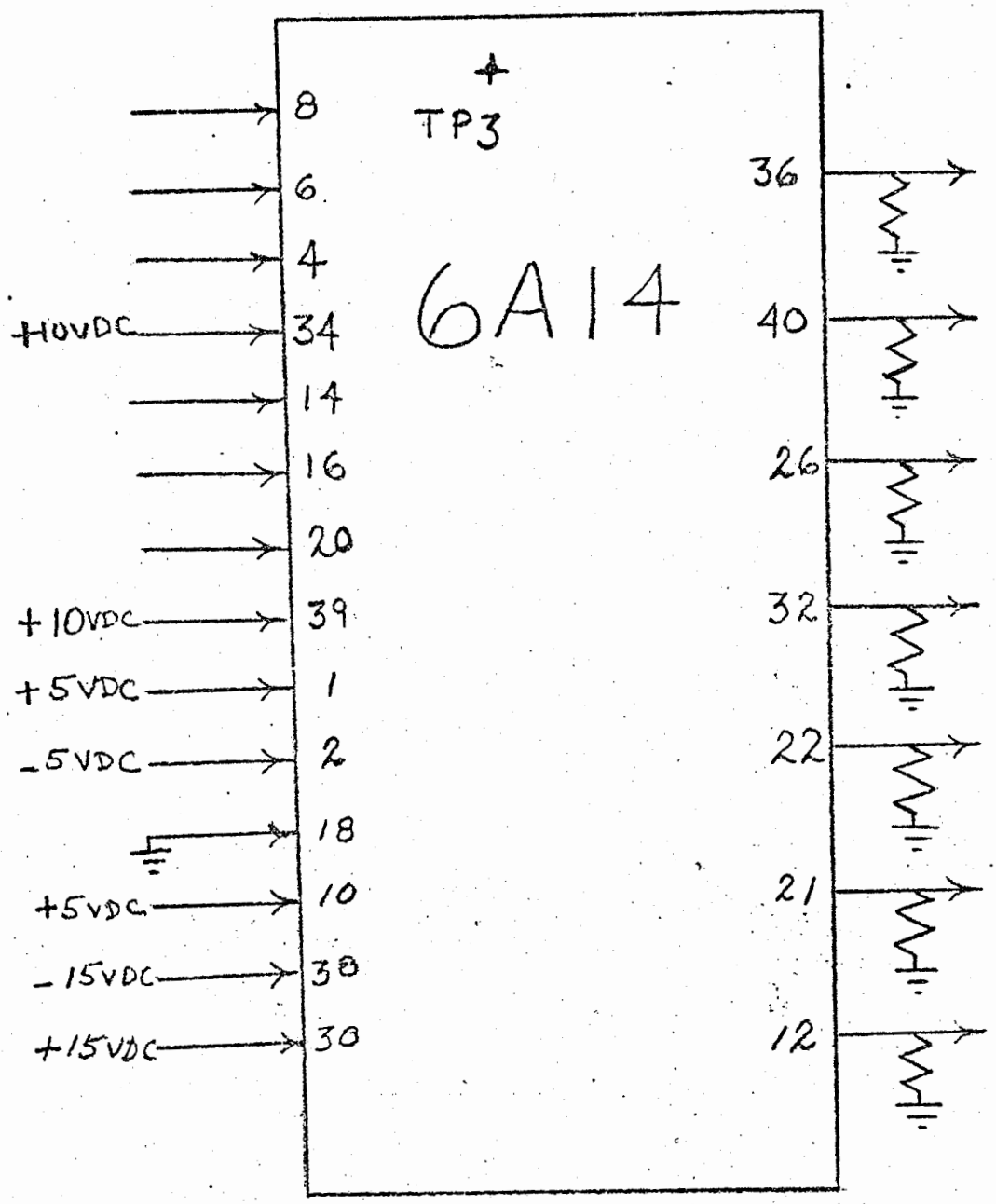


Figure 1

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TABLE II

	INPUT(BCD)			OUTPUT(VDC)						
PINS	20	14	16	36	40	26	32	22	21	12
	0	0	0	-0.828±0.600	———	-0.828±0.600	———	14.88±0.500	———	-2.400±0.500
	0	0	0	-1.660±0.750	———	-1.660±0.750	———	7.51±0.500	———	-1.210±0.500
	0	1	0	-2.500±0.100	———	-2.50±0.100	———	4.96±0.500	———	-0.806±0.500
	0	1	1	-3.330±0.120	———	-3.330±0.120	———	3.72±0.500	———	-0.606±0.500
	1	0	0	-5.000±0.120	———	-5.000±0.120	———	2.48±0.500	———	-0.401±0.500
	1	0	1	-6.660±0.130	———	-6.660±0.130	———	1.68±0.500	———	-0.301±0.500
	1	1	0	-8.280±0.150	———	-8.28±0.150	———	1.49±0.500	———	-0.243±0.500
	1	1	1	-10.020±0.175	-3.336±0.175	-10.02±0.175	-5.360±0.200	1.24±0.500	0.487±0.500	-0.200±0.500

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4.2.2.5 Multiplex Switch Output.- With BCD logic applied to the inputs as shown in Tables I and II the corresponding output shall be an incremental voltages when measured across a 10K dummy load resistor. A BCD code of "0ll" shall be applied to pin 4, 6 and 8 respectively in order to obtain -1.24 ± 0.10 VDC at TP3 throughout the performance test.

TABLE I

	Input (BCD)			Output (VDC)
Pins	4	6	8	Test Point 3
	0	0	0	5.000 ± 0.100
	0	0	1	3.500 ± 0.100
	0	1	0	1.660 ± 0.100
	0	1	1	-1.240 ± 0.100
	1	0	0	-1.670 ± 0.100
	1	0	1	-2.670 ± 0.100
	1	1	0	-3.500 ± 0.100
	1	1	1	-5.000 ± 0.100

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746273.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicalbe to this specification.

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Preparing Activity:

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Project No. 1240-A607

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